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Developing device and image forming apparatus

Abstract

The developing device includes a development container, a developer carrier, and a toner trapping mechanism. The development container includes a first conveyance chamber and a second conveyance chamber placed in parallel juxtaposition and communicated with each other at their longitudinal both end portions, and the development container contains a toner-containing developer to be fed to an image carrier. The toner trapping mechanism includes a duct, an air inlet port, a trapping filter, and a fan. The duct is connected to the second conveyance chamber, and allows air present in the second conveyance chamber to flow therethrough. The development container has an air vent opened in the first conveyance chamber and communicated with outside of the development container.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based on and claims the benefit of priority from Japanese Patent Application No. 2022-191115 filed on Nov. 30, 2022, the contents of which are hereby incorporated by

reference.

BACKGROUND

(2) The present disclosure relates to a developing device and an image forming apparatus.

(3) In an image forming apparatus of electrophotographic system such as copiers and printers, there has been widely used a device for forming a toner image, which is to be transferred onto a paper sheet in later process, by feeding toner and executing development for an electrostatic latent image formed on an outer circumferential surface of a photosensitive drum or other image carrier. In order to enable continuous formation of uniform images, the image forming apparatus conveys toner-containing developer contained in a development container while stirring the developer in the development container.

(4) With conventional image forming apparatuses, there has been a fear that toner may scatter from inside to outside of the development container, causing apparatus inside to be contaminated with the scattered toner.

SUMMARY

(5) A developing device according to one aspect of the present disclosure comprises a development container, a first conveyance member and a second conveyance member, a developer carrier, and a toner trapping mechanism. The development container includes a first conveyance chamber and a second conveyance chamber placed in parallel juxtaposition and communicated with each other at their longitudinal both end portions, and the development container contains a toner-containing developer to be fed to an image carrier. The first conveyance member and the second conveyance member are rotatably supported by the first conveyance chamber and the second conveyance chamber, respectively, and convey and circulate the developer, while stirring the developer, in mutually opposite directions of the longitudinal direction. The developer carrier is rotatably supported by the development container in opposition to the image carrier, and feeds the toner contained in the second conveyance chamber to the image carrier. The toner trapping mechanism traps the toner contained in the second conveyance chamber. The toner trapping mechanism includes a duct, an air inlet port, a trapping filter, and a fan. The duct is connected to the second conveyance chamber and allows air present in the second conveyance chamber to flow therethrough. The air inlet port is opened at a connecting portion between the duct and the second conveyance chamber so as to allow air present in the second conveyance chamber to flow into the duct. The trapping filter traps the toner contained in air flowing through the duct. The fan allows air present in the second conveyance chamber to be sucked into the duct and let to flow outside. The development container has an air vent which is opened in the first conveyance chamber and communicated with outside of the development container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic cross-sectional front view of an image forming apparatus according to one embodiment of the present disclosure;

(2) FIG. **2** is a block diagram showing a configuration of the image forming apparatus of FIG. **1**;

(3) FIG. **3** is a schematic cross-sectional front view of around an image forming part in the image forming apparatus of FIG. **1**;

(4) FIG. **4** is a vertical cross-sectional front view of a developing device in the image forming part of FIG. **3**;

(5) FIG. **5** is a horizontal cross-sectional plan view of the developing device in the image forming part of FIG. **3**;

(6) FIG. **6** is a vertical cross-sectional side view of the developing device in the image forming part of FIG. **3**;

(7) FIG. **7** is a horizontal cross-sectional plan view of a developing device according to Modification Example 1;
(8) FIG. **8** is a horizontal cross-sectional plan view of a developing device according to Modification Example 2;
(9) FIG. **9** is a vertical cross-sectional side view of the developing device according to Modification Example 2; and
(10) FIG. **10** is a vertical cross-sectional front view of a developing device according to Modification Example 3.

DETAILED DESCRIPTION

(11) Hereinbelow, an embodiment of the present disclosure will be described with reference to the accompanying drawings. It is to be noted that the present disclosure is not limited to the following contents.

(12) FIG. **1** is a schematic cross-sectional front view of an image forming apparatus **1** according to this embodiment. FIG. **2** is a block diagram showing a configuration of the image forming apparatus **1** of FIG. **1**. FIG. **3** is a schematic cross-sectional front view of around an image forming part **20** in the image forming apparatus **1** of FIG. **1**. An example of the image forming apparatus **1** of the embodiment is a tandem-type color printer in which a toner image is transferred onto a paper sheet S by using an intermediate transfer belt **31**. The image forming apparatus **1** may also be a so-called multifunction peripheral equipped with such functions as printing, scanning (image reading), and facsimile transmission.

(13) As shown in FIGS. **1**, **2** and **3**, the image forming apparatus **1** includes a sheet feed part **3**, a sheet conveyance part **4**, an exposure part **5**, image forming parts **20**, a transfer part **30**, a fixing part **6**, a sheet discharge part **7**, and a controller **8**, as these are provided in an apparatus housing **2**.

(14) The sheet feed part **3** is placed at a bottom portion of the housing **2**. The sheet feed part **3**, containing a plurality of unprinted paper sheets S, separates a sheet S therefrom and feeds out the sheet S one by one on occasions of printing. The sheet conveyance part **4** extends in an up/down direction along a side wall of the housing **2**. The sheet conveyance part **4** conveys the sheet S, which has been fed out from the sheet feed part **3**, to a secondary transfer part **33** and the fixing part **6**, and further discharges the after-fixation sheet S through a sheet discharge port **4***a* to the sheet discharge part **7**. The exposure part **5** is placed above the sheet feed part **3**. The exposure part **5** applies laser light, which has been controlled based on image data, toward the image forming parts **20**.

(15) The image forming parts **20** are placed above the exposure part **5** and below the intermediate transfer belt **31**. The image forming parts **20** include a yellow-destined image forming part **20**Y, a cyan-destined image forming part **20**C, a magenta-destined image forming part **20**M, and a black-destined image forming part **20**B. These four image forming parts **20** are identical in basic configuration. Therefore, hereinafter, unless otherwise necessarily particularly limited, the identification signs ‘Y’, ‘C’, ‘M’ and ‘B’ representing individual colors, respectively, may be omitted from time to time.

(16) Each image forming part **20** includes a photosensitive drum (image carrier) **21** which is supported rotatable in a specified direction (clockwise in FIGS. **1** and **3**). The image forming part **20** further includes a charging part **22**, a developing device **40**, and a drum cleaning part (cleaning part) **23**, as these are disposed around the photosensitive drum **21** along its rotational direction. In addition, a primary transfer part **32** is placed between the developing device **40** and the drum cleaning part **23**.

(17) The photosensitive drum **21**, which is formed into a horizontally-extending cylindrical shape, has, on its outer circumferential surface, a photosensitive layer formed from amorphous silicon photoconductor, as an example. The charging part **22** electrically charges the surface (outer circumferential surface) of the photosensitive drum **21** to a specified potential. The exposure part **5** illuminates the outer circumferential surface of the photosensitive drum **21** charged by the charging

part **22** so that an electrostatic latent image of an original image is formed on the outer circumferential surface of the photosensitive drum **21**. The developing device **40** feeds toner to the electrostatic latent image and makes development to form a toner image. The four image forming parts **20** form toner images of different colors, respectively. After a toner image is primarily transferred onto an outer circumferential surface of the intermediate transfer belt **31**, the drum cleaning part **23** removes toner and the like remaining on the outer circumferential surface of the photosensitive drum **21**, thus fulfilling cleaning function. In this way, the image forming part **20** forms an image (toner image) that is to be transferred onto the sheet S in later process.
(18) The transfer part **30** includes the intermediate transfer belt **31**, primary transfer parts **32**Y, **32**C, **32**M, **32**B, a secondary transfer part **33**, and a belt cleaning part **34**. The intermediate transfer belt **31** is placed above the four image forming parts **20**. The intermediate transfer belt **31** is an endless intermediate transferer which is supported so as to be turnable in a specified direction (counterclockwise in FIG. **1**) and to which toner images formed by the four image forming parts **20**, respectively, are primarily transferred in superimposition one after another. The four image forming parts **20** are placed in such a so-called tandem mode as to be arrayed in one line from upstream side toward downstream side of the turning direction of the intermediate transfer belt **31**.
(19) The primary transfer parts **32**Y, **32**C, **32**M, **32**B are placed upward of the individual-color image forming parts **20**Y, **20**C, **20**M, **20**B, respectively, with the intermediate transfer belt **31** pinched therebetween. The secondary transfer part **33** is placed upstream of the fixing part **6** as viewed in a sheet conveyance direction of the sheet conveyance part **4**, as well as downstream of the four image forming parts **20**Y, **20**C, **20**M, **20**B as viewed in the turning direction of the intermediate transfer belt **31**. The belt cleaning part **34** is placed downstream of the secondary transfer part **33** as viewed in the turning direction of the intermediate transfer belt **31**.
(20) The primary transfer part **32** transfers toner images, which have been formed on the outer circumferential surfaces of the photosensitive drums **21**, onto the intermediate transfer belt **31**. In other words, the toner images are primarily transferred onto the outer circumferential surface of the intermediate transfer belt **31** at the individual-color primary transfer parts **32**Y, **32**C, **32**M, **32**B, respectively. Thus, by the toner images of the four image forming parts **20** being transferred successively in superimposition at specified timings along with the turning of the intermediate transfer belt **31**, a color toner image in which four-color toner images of yellow, cyan, magenta and black have been superimposed together is formed on the outer circumferential surface of the intermediate transfer belt **31**.
(21) The color toner image on the outer circumferential surface of the intermediate transfer belt **31** is transferred onto the sheet S fed in synchronization by the sheet conveyance part **4** at a secondary transfer nip portion formed in the secondary transfer part **33**. The belt cleaning part **34** removes toner and other deposits remaining on the outer circumferential surface of the intermediate transfer belt **31** after secondary transfer, fulfilling the cleaning function. In this way, the transfer part **30** transfers (records) the toner image, which has been formed on the outer circumferential surface of the photosensitive drum **21**, onto the sheet S.
(22) The fixing part **6** is placed upward of the secondary transfer part **33**. The fixing part **6** heats and pressurizes the sheet S, onto which the toner image has been transferred, so as to fix the toner image on the sheet S.
(23) The sheet discharge part **7** is placed above the transfer part **30**. The sheet S, on which the toner image has been fixed and for which printing is over, is conveyed to the sheet discharge part **7**. The sheet discharge part **7** allows an after-printing sheet (printed matter) to be taken out from upward.
(24) The controller **8** includes a CPU, an image processing part, a storage part, and other electronic circuits and electronic components (none shown). The CPU, based on control programs and data stored in the storage part, controls operations of the individual component elements provided in the image forming apparatus **1** to execute processing related to functions of the image forming apparatus **1**. The sheet feed part **3**, the sheet conveyance part **4**, the exposure part **5**, the image

forming parts **20**, the transfer part **30** and the fixing part **6** receive instructions individually from the controller **8** to fulfill printing on the sheet S in linkage with one another. The storage part is made up by a combination of nonvolatile storage device such as program ROM (Read Only Memory), data ROM, or the like and volatile storage device such as RAM (Random Access Memory).
(25) Also, the image forming apparatus **1**, as shown in FIG. **2**, further includes a voltage application part **12** and a current detection part **13**.
(26) The voltage application part **12** includes a power supply part and a control circuit, as an example (neither shown). The voltage application part **12** is electrically connected to a later-described developing roller (developer carrier) **44** of the developing device **40**. The voltage application part **12** applies a developing voltage (developing bias) to the developing roller **44**. Via the voltage application part **12**, the controller **8** controls application timing, voltage value, polarity, application time or the like of the developing voltage applied to the developing roller **44**.
(27) Under application of the developing voltage to the developing roller **44**, the current detection part **13** detects an electric current flowing between the photosensitive drum **21** and the developing roller **44**. From the current detection part **13**, the controller **8** receives information as to the current detected by the current detection part **13**.
(28) Next, a configuration of the developing device **40** will be described with reference to FIGS. **4**, **5** and **6** in addition to FIGS. **2** and **3**. FIGS. **4**, **5** and **6** are a vertical cross-sectional front view, a horizontal cross-sectional plan view, and a vertical cross-sectional side view of the developing device **40** in the image forming part **20** of FIG. **3**. In addition, the individual-color developing devices **40** are identical in basic configuration and, therefore, expression and explanation of identification signs representing the individual colors with respect to the component elements are omitted. Also, in the following description, the terms ‘axial direction’ refers to an axial direction (drawing-sheet depthwise direction in FIGS. **3** and **4**, left/right lateral direction in FIGS. **5** and **6**) of each rotation of the photosensitive drum **21**, a first conveyance member **42**, a second conveyance member **43** and the developing roller **44**, all of which extend parallel to one another.
(29) The developing device **40** feeds toner to the outer circumferential surface of the photosensitive drum **21**. The developing device **40** is settable to and removable from the housing **2** of the image forming apparatus **1**, as an example. The developing device **40** includes a development container **50**, a first conveyance member **42**, a second conveyance member **43**, a developing roller (developer carrier) **44**, and a restricting blade **45**.
(30) The development container **50**, having a slender shape extending along the axial direction of the photosensitive drum **21**, is placed with its longitudinal direction positioned horizontal. That is, the longitudinal direction of the development container **50** is parallel to the axial direction of the photosensitive drum **21**. The development container **50** contains, for example, a two-component developer containing toner and magnetic carrier as the toner-containing developer to be fed to the photosensitive drum **21**.
(31) The development container **50** includes a partitioning portion **51**, a first conveyance chamber **52**, a second conveyance chamber **53**, a first communicating portion **54**, and a second communicating portion **55**.
(32) The partitioning portion **51** is provided in lower part inside the development container **50**. The partitioning portion **51** is placed at a generally central portion in a direction intersecting the longitudinal direction of the development container **50** (left/right lateral direction in FIG. **4**, up/down direction in FIG. **5**). The partitioning portion **51** is formed into a generally plate shape extending in the longitudinal direction of the development container **50** as well as in the up/down direction. The partitioning portion **51** partitions interior of the development container **50** in a direction intersecting the longitudinal direction.
(33) The first conveyance chamber **52** and the second conveyance chamber **53** are provided inside the development container **50**. The first conveyance chamber **52** and the second conveyance chamber **53** are formed by the interior of the development container **50** being partitioned by the

partitioning portion **51**. The first conveyance chamber **52** and the second conveyance chamber **53** are juxtaposed generally equal in height to each other.
(34) The second conveyance chamber **53**, including a vicinity of a placement area of the developing roller **44** inside the development container **50**, is placed in adjacency to the photosensitive drum **21**. The first conveyance chamber **52** is placed in a region in the development container **50** isolated from the photosensitive drum **21** more than the second conveyance chamber **53**. The first conveyance chamber **52**, to which a developer supply pipe (not shown) is connected, is fed with the developer via the developer supply pipe. In the first conveyance chamber **52**, the developer is conveyed in a first direction f**1** by the first conveyance member **42**. In the second conveyance chamber **53**, the developer is conveyed by the second conveyance member **43** in a second direction f**2** opposite to the first direction f**1**.
(35) The first communicating portion **54** and the second communicating portion **55** are placed outside both end portions, respectively, of the partitioning portion **51** in its longitudinal direction. The first communicating portion **54** and the second communicating portion **55** allow the first conveyance chamber **52** and the second conveyance chamber **53** to be communicated with each other in a direction intersecting the longitudinal direction of the partitioning portion **51** (left/right lateral direction in FIG. **4**, up/down direction in FIG. **5**), i.e., in a thicknesswise direction of the generally plate-shaped partitioning portion **51**. In other words, the first communicating portion **54** and the second communicating portion **55** allow the first conveyance chamber **52** and the second conveyance chamber **53** to be communicated with each other at their longitudinal-direction both end-side portions.
(36) The first communicating portion **54** allows a first-direction-f**1** downstream end of the first conveyance chamber **52** and a second-direction-f**2** upstream end of the second conveyance chamber **53** to be communicated with each other. In the first communicating portion **54**, the developer is conveyed from the first conveyance chamber **52** side toward the second conveyance chamber **53** side. The second communicating portion **55** allows a second-direction-f**2** downstream end of the second conveyance chamber **53** and a first-direction-f**1** upstream end of the first conveyance chamber **52** to be communicated with each other. In the second communicating portion **55**, the developer is conveyed from the second conveyance chamber **53** side toward the first conveyance chamber **52** side.
(37) The first conveyance member **42** is placed inside the first conveyance chamber **52**. The second conveyance member **43** is placed inside the second conveyance chamber **53**. The first conveyance member **42** and the second conveyance member **43** are juxtaposed generally equal in height to each other. The second conveyance member **43** extends in proximity and parallel to the developing roller **44**. The first conveyance member **42** and the second conveyance member **43** are supported by the development container **50** so as to be rotatable about an axis extending horizontally in parallel to the developing roller **44**.
(38) The first conveyance member **42** and the second conveyance member **43** are identical in basic configuration to each other. The first conveyance member **42** has helical-shaped first conveyance vanes **42***b* at an outer circumferential portion of a rotating shaft **42***a* extending along the longitudinal direction of the development container **50**. The second conveyance member **43** has helical-shaped second conveyance vanes **43***b* at an outer circumferential portion of a rotating shaft **43***a* extending along the longitudinal direction of the development container **50**.
(39) Within the first conveyance chamber **52**, the first conveyance member **42**, while stirring the developer, conveys the developer in the first direction f**1** directed from the second communicating portion **55** side toward the first communicating portion **54** side along the rotational axial direction. Within the second conveyance chamber **53**, the second conveyance member **43**, while stirring the developer, conveys the developer in the second direction f**2** from the first communicating portion **54** side toward the second communicating portion **55** side along the rotational axial direction. That is, the first conveyance member **42** and the second conveyance member **43**, while stirring, conveys

the developer in mutually opposite directions of the longitudinal direction, thus circulating the developer in a specified circulative direction.

(40) The developing roller **44** is positioned above the second conveyance member **43** inside the development container **50** and also placed in opposition to the photosensitive drum **21**. The developing roller **44** is supported by the development container **50** so as to be rotatable about an axis extending parallel to the axis of the photosensitive drum **21**. The developing roller **44** has a cylindrical-shaped developing sleeve **441** that rotates counterclockwise during image formation, for example as in FIGS. **3** and **4**, and a stationary magnet **442** unrotatably fixed within the developing sleeve (see FIG. **4**).

(41) The developing roller **44** has its outer circumferential surface partly exposed from the development container **50** and is set into proximate opposition to the photosensitive drum **21**. The developing roller **44** carries toner on part of its outer circumferential surface, where the toner is to be fed to the outer circumferential surface of the photosensitive drum **21**, in a region opposed to the photosensitive drum **21**. The developing roller **44** carries toner within the second conveyance chamber **53** of the development container **50**, and feeds the toner to the photosensitive drum **21**. In other words, the developing roller **44** makes toner in the second conveyance chamber **53** deposited to an electrostatic latent image on the outer circumferential surface of the photosensitive drum **21**, thus forming a toner image.

(42) The restricting blade **45** is placed upstream of an oppositional region between the developing roller **44** and the photosensitive drum **21** as viewed in a rotational direction of the developing roller **44**. The restricting blade **45** is placed in proximate opposition to the developing roller **44** with a specified clearance provided between a forward end of the restricting blade **45** and the outer circumferential surface of the developing roller **44**. The restricting blade **45** extends over an entire axial length of the developing roller **44**. The restricting blade **45** restricts a layer thickness of the developer carried by the outer circumferential surface of the developing roller **44** that passes through the clearance between the forward end of the restricting blade **45** and the outer circumferential surface of the developing roller **44**.

(43) Along with rotations of the first conveyance member **42** and the second conveyance member **43**, developer in the development container **50** circulates in a specified circulating direction between the first conveyance chamber **52** and the second conveyance chamber **53** via the first communicating portion **54** and the second communicating portion **55**. In this case, the toner in the development container **50** is stirred and electrically charged so as to be carried on the outer circumferential surface of the developing roller **44**. The developer carried on the outer circumferential surface of the developing roller **44**, after restricted in its layer thickness by the restricting blade **45**, is conveyed to the oppositional region between the developing roller **44** and the photosensitive drum **21** by the rotation of the developing roller **44**. When a specified developing voltage is applied to the developing roller **44**, toner in the developer carried on the outer circumferential surface of the developing roller **44** is moved to the outer circumferential surface of the photosensitive drum **21** in the oppositional region due to a potential difference from a potential of the surface (outer circumferential surface) of the photosensitive drum **21**. As a result of this, the electrostatic latent image on the outer circumferential surface of the photosensitive drum **21** is developed by the toner.

(44) Next, a further detailed configuration of the developing device **40** will be described below with reference to FIGS. **4**, **5** and **6**. It is noted that arrows representing an airflow direction fd in a duct **61** are added in FIGS. **4** and **6**.

(45) The developing device **40** includes a toner trapping mechanism **60**. The toner trapping mechanism **60** traps toner present in the second conveyance chamber **53**. The toner trapping mechanism **60** includes the duct **61**, a trapping filter **62**, and a fan **63**.

(46) The duct **61** is placed in adjacency to the second conveyance chamber **53**. The duct **61** is opposed to the photosensitive drum **21** with a placement area of the developing roller **44** in the

development container **50** interposed therebetween as viewed in a direction intersecting the longitudinal direction of the development container **50** (left/right lateral direction in FIG. **4**, drawing-sheet depthwise direction in FIG. **6**). The duct **61** is connected to the second conveyance chamber **53** at an upstream end in the airflow direction. The duct **61** allows air within the second conveyance chamber **53** to be circulated therethrough. The duct **61** has an air inlet port **611** and an air exhaust port **612**.

(47) The air inlet port **611**, which is a connecting portion of the duct **61** with the second conveyance chamber **53**, is placed above the developing roller **44**. The air inlet port **611** is positioned at an upstream end of the duct **61** in the airflow direction. The air inlet port **611** is opened over an entire longitudinal length of the second conveyance chamber **53**. The air inlet port **611** is formed into, for example, a rectangular shape elongated in the longitudinal direction of the second conveyance chamber **53**, and opposed to the developing roller **44**. The air inlet port **611** allows interior of the second conveyance chamber **53** and interior of the duct **61** to be communicated with each other. Air in the second conveyance chamber **53** is permitted to flow into the duct **61** through the air inlet port **611**.

(48) The air exhaust port **612** is placed, for example, at a back portion of the development container **50**. The air exhaust port **612** is placed at a downstream end of the duct **61** in its airflow direction. The air exhaust port **612** permits air present in the duct **61** to flow out. That is, air in the second conveyance chamber **53** is permitted to flow out from within the duct **61** through the air exhaust port **612**.

(49) The trapping filter **62** is placed in the duct **61**. The trapping filter **62** covers an airflow cross-section of the duct **61**. That is, air in the second conveyance chamber **53** that has flowed into the duct **61** through the air inlet port **611** is permitted to pass through the trapping filter **62**. As a result of this, the trapping filter **62** traps toner contained in airflows passing through the duct **61**.

(50) The fan **63** is connected to the air exhaust port **612** of the duct **61**. As the fan **63** is driven (rotated forward), air in the second conveyance chamber **53** is forcedly sucked into the duct **61** and then let to flow outside through the air exhaust port **612**. In other words, the fan **63** forcedly sucks air in the second conveyance chamber **53** into the duct **61**, letting the air flow outside.

(51) As shown in FIG. **6**, an exhaust side of the fan **63** under its forward rotation is connected to a housing duct **2***a* provided in the housing **2** of the image forming apparatus **1**. The housing duct **2***a* extends up to an air exhaust part (not shown) provided in exterior of the image forming apparatus **1** and communicated with outside of the image forming apparatus **1**. As a result of this, the fan **63** allows the air in the second conveyance chamber **53** to flow outside of the image forming apparatus **1** via the duct **61** and the housing duct **2***a*. Otherwise, the fan **63** may be placed in the exhaust part to which the housing duct **2***a* is connected so that the exhaust part is communicated with outside of the image forming apparatus **1**.

(52) Then, as shown in FIGS. **4** and **5**, the development container **50** has an air vent **56** opened in the first conveyance chamber **52**. In this embodiment, as an example, the air vent **56** is formed into a rectangular shape extending along the longitudinal direction of the first conveyance chamber **52** and placed at a longitudinal central portion of the first conveyance chamber **52**. In addition, the air vent **56** may also be formed over a longitudinal entire length of the first conveyance chamber **52**.

(53) The air vent **56** extends inward and outward through a peripheral wall portion of the first conveyance chamber **52**. That is, the air vent **56** is communicated with outside of the development container **50**. As a result of this, the air vent **56** permits air within the first conveyance chamber **52** to flow outward of the development container **50**.

(54) According to the above-described configuration, since air within the first conveyance chamber **52** is permitted to flow outward of the development container **50** through the air vent **56**, any pressure increases within the first conveyance chamber **52** can be suppressed. As a consequence, toner present in the stirring region in the developing device **40** can be prevented from flying-and-rising and reaching the toner trapping mechanism **60**. That is, trapping of more than necessary

quantity of toner by the trapping filter **62** can be suppressed, making it implementable to enhance stability of the toner trapping mechanism **60** in its trapping performance. Consequently, in the developing device **40**, it becomes possible to continue the stable trapping of scattered toner.

(55) Also, the air vent **56** is formed at an upper portion of the first conveyance chamber **52**. In more detail, the air vent **56** is formed at a ceiling portion of the first conveyance chamber **52** so as to extend in an up/down direction through the peripheral wall portion of the first conveyance chamber **52**. That is, since the air vent **56** is opened upward relative to the first conveyance chamber **52**, the developer can be prevented from unintentionally leaking downward via the air vent **56**.

(56) Also, the development container **50** has a ventilation filter **57** covering the air vent **56**. In more detail, the ventilation filter **57** covers an airflow cross section of the air vent **56**. That is, air that flows out from within the first conveyance chamber **52** via the air vent **56** outward of the development container **50** passes through the ventilation filter **57**. As a result of this, the ventilation filter **57** traps toner contained in the air passing through the air vent **56**. Thus, it becomes possible to enhance the effect of preventing toner from leaking via the air vent **56**.

(57) In addition, placement of the ventilation filter **57** may be omitted, for example, by providing a ventilation duct which is formed into a cylindrical shape extending continuously upward from the air vent **56**.

(58) FIG. **7** is a horizontal cross-sectional plan view of a developing device **40** according to Modification Example 1. In the developing device **40** of Modification Example 1, the development container **50** has two air vents **56**A, **56**B. The air vents **56**A, **56**B are placed at both upstream-and-downstream end portions, respectively, in the developer conveyance direction (first direction f**1**) of the first conveyance chamber **52**. At both upstream-and-downstream end portions of the developer conveyance direction of the first conveyance chamber **52**, the first conveyance chamber **52** is communicated with the second conveyance chamber **53**, making the first conveyance chamber **52** more easily increasable in pressure. Accordingly, providing the air vents **56**A, **56**B makes it possible to suppress pressure increases by virtue of outflow of air that is permitted to flow outward via the air vents **56**A, **56**B at both end portions of the first conveyance chamber **52**.

(59) In addition, the air vent may be placed at at least one downstream-side end portion out of upstream-and-downstream both end portions as viewed in the developer conveyance direction (first direction f**1**) of the first conveyance chamber **52**. As a result of this, with regard to the first conveyance chamber **52**, air can be let to flow outward via the air vent **56**B at the downstream-side end portion of the first conveyance chamber **52**, at which in particular pressure increases are more likely to occur. Accordingly, pressure increases within the first conveyance chamber **52** can be suppressed.

(60) FIGS. **8** and **9** are a horizontal cross-sectional plan view and a vertical cross-sectional side view, respectively, of the developing device **40** according to Modification Example 2. In the developing device **40** of Modification Example 2, the development container **50** includes a discharge part **58**. The discharge part **58** is provided further downstream of the downstream end of the second conveyance chamber **53** in the second direction f**2**. The discharge part **58** connects to the second conveyance chamber **53**. Interior of the discharge part **58** and the interior of the second conveyance chamber **53** are communicated with each other. The discharge part **58** is smaller in inner diameter than the second conveyance chamber **53**. The discharge part **58** has a developer discharge port **581**.

(61) In addition, the rotating shaft **43***a* of the second conveyance member **43** extends continuously to within the discharge part **58**. One axial end of the rotating shaft **43***a* is rotatably supported by the development container **50** at a downstream end of the discharge part **58** as viewed in the second direction f**2** of the second conveyance chamber **53**.

(62) The developer discharge port **581** is placed at a downstream end of the discharge part **58** as viewed in the second direction f**2** of the second conveyance chamber **53**. For example, the developer discharge port **581** is opened under the rotating shaft **43***a* of the second conveyance

member **43**. The developer discharge port **581** allows excess developer within the development container **50** to be discharged outside.

(63) The second conveyance member **43** further includes restricting vanes **43***c* and discharge vanes **43***d* in addition to the second conveyance vanes **43***b*. These three kinds of vanes are provided in an order of the second conveyance vanes **43***b*, the restricting vanes **43***c* and the discharge vanes **43***d* as mentioned from the second conveyance chamber **53** side toward the discharge part **58**. The restricting vanes **43***c* and the discharge vanes **43***d* are provided integrally with the rotating shaft **43***a*, like the second conveyance vanes **43***b*, so as to extend helically along the axial direction of the rotating shaft **43***a* in its outer circumferential portion.

(64) The restricting vanes **43***c* are positioned downstream of the second conveyance vanes **43***b* as viewed in the second direction f**2** of the second conveyance chamber **53**, and placed within the second conveyance chamber **53**. The restricting vanes **43***c* are opposed to a connecting portion between the second conveyance chamber **53** and the discharge part **58** in the axial direction of the rotating shaft **43***a*.

(65) The restricting vanes **43***c* are inverse to the second conveyance vanes **43***b* in terms of winding direction. As a result of this, the restricting vanes **43***c* block the developer that has been conveyed to near the downstream end within the second conveyance chamber **53**, so that movement of the developer to the discharge part **58** side is restricted. It is noted that the restricting vanes **43***c* are smaller in pitch than the second conveyance vanes **43***b*.

(66) An outer circumferential portion of the restricting vanes **43***c* has a specified clearance against an inner surface of the development container **50**. Developer having exceeded a specified quantity in the second conveyance chamber **53** is conveyed, as excess developer, through the clearance between the outer circumferential portion of the restricting vanes **43***c* and the inner surface of the development container **50** toward the discharge part **58**.

(67) The discharge vanes **43***d* are positioned downstream of the restricting vanes **43***c* as viewed in the second direction f**2** of the second conveyance chamber **53**, and placed within the discharge part **58**. The discharge vanes **43***d* are identical in winding direction to the second conveyance vanes **43***b*. That is, the developer conveyance direction in the discharge part **58** is identical to the second direction f**2** of the second conveyance chamber **53**. As a result of this, the discharge vanes **43***d* convey excess developer in the discharge part **58** toward the developer discharge port **581**. It is noted that an outer diameter of the discharge vanes **43***d* is smaller than outer diameters of the second conveyance vanes **43***b* and the restricting vanes **43***c*. A pitch of the discharge vanes **43***d* is smaller than a pitch of the second conveyance vanes **43***b*.

(68) The development container **50** has the air vent **56**. The air vent **56** is placed at a downstream-side end portion of the first conveyance chamber **52** as viewed in its developer conveyance direction (first direction f**1**). Moreover, the air vent **56** is placed at an opposite-side end portion of the first conveyance chamber **52** in its developer conveyance direction, which is opposite to the side on which the developer discharge port **581** is placed.

(69) According to the above-described configuration, outflow of air present within the development container **50** through the developer discharge port **581** can be expected on the upstream side of the first conveyance chamber **52** in its developer conveyance direction. That is, the developer discharge port **581** can be made to act for suppression of pressure increases. Furthermore, providing the air vent **56** also on the downstream side of the first conveyance chamber **52** in its developer conveyance direction contributes to suppressing pressure increases within the first conveyance chamber **52**.

(70) Developer to be used for formation of toner images in the image forming apparatus **1** is a two-component developer containing magnetic carrier and toner. With a two-component developer, it is known that toner scattering from the development container **50** is more likely to occur. Therefore, in the image forming apparatus **1** with the use of a two-component developer, providing the air vent **56** as described above makes it possible to even more effectively suppress toner scattering in the

image forming apparatus **1**.

(71) Reverting to FIG. **4**, the trapping filter **62** includes a first trapping filter **621** and a second trapping filter **622**. The first trapping filter **621** and the second trapping filter **622** are both retained by a single retaining member **64**. The retaining member **64** is placed at an airflow-direction upstream portion in the duct **61** so as to be adjacent to the air inlet port **611**.

(72) The first trapping filter **621** is placed at a site of the air inlet port **611**, which is a connecting portion between the duct **61** and the second conveyance chamber **53**. The first trapping filter **621**, identical in shape to the air inlet port **611**, is formed into, for example, a rectangular shape elongated in the longitudinal direction of the second conveyance chamber **53**. The first trapping filter **621** covers the air inlet port **611**. That is, the first trapping filter **621** is opposed to the developing roller **44**. The first trapping filter **621** traps toner contained in air that flows from the second conveyance chamber **53** into the duct **61**.

(73) The second trapping filter **622** is placed on a downstream side of the first trapping filter **621** as viewed in the airflow direction in the duct **61**. The second trapping filter **622**, which is identical in shape to a cross section in a direction intersecting the airflow direction in the duct **61**, is formed into, for example, a rectangular shape elongated in the longitudinal direction of the second conveyance chamber **53**. The second trapping filter **622** covers the airflow cross-section in the duct **61**. The second trapping filter **622** traps toner contained in the air passing through the first trapping filter **621** and flowing within the duct **61**.

(74) The first trapping filter **621** is a nonwoven fabric which is made from, for example, circular-in-cross-section fabric having an outer diameter of 10 to 20 μm and which has a thickness of about 1 mm. The second trapping filter **622** is a nonwoven fabric which is made from, for example, circular-in-cross-section fabric having an outer diameter of 20 to 40 μm and which has a thickness of about 0.2 mm. The second trapping filter **622** is enhanced in toner trapping efficiency by virtue of its being made finer in mesh roughness than the first trapping filter **621**.

(75) According to the makeup of the trapping filter **62**, the first trapping filter **621** can be kept from large-quantity trapping of scattered toner present in the second conveyance chamber **53** (around the developing roller **44**), thus being made less likely to be clogged. Further, the second trapping filter **622** makes it possible to prevent leakage of toner outside of the development container **50**. Furthermore, setting the first trapping filter **621** larger in opening area than the second trapping filter **622** makes it implementable to uniformize sucking force by the fan **63** at the site of the first trapping filter **621**.

(76) As shown in FIGS. **2**, **6** and **9**, the image forming apparatus **1** further includes a vibration generating part **14**. The vibration generating part **14** is set on the housing **2** of the image forming apparatus **1**.

(77) The vibration generating part **14** is placed in opposition to the development container **50** outside the developing device **40**. The vibration generating part **14** includes, for example, a vibrating motor, a control board, as well as other electronic circuits and electronic components (none shown). On an output shaft of the vibrating motor, attached is an exciting weight whose center of gravity is eccentric to a rotational axis of the output shaft of the vibrating motor.

(78) The vibration generating part **14** is connected to the retaining member **64** via a connecting member **14***a*. As the vibrating motor is driven, the vibration generating part **14** makes the trapping filter **62** vibrated via the retaining member **64**. In other words, vibrations are imparted to the trapping filter **62** by the vibration generating part **14**.

(79) By the first trapping filter **621** being vibrated by the vibration generating part **14**, toner trapped by the first trapping filter **621** and deposited on the first trapping filter **621** can be dropped into the second conveyance chamber **53**. Therefore, functions of the first trapping filter **621** can be recovered, making it possible to continuously suppress toner scattering in the image forming apparatus **1**. It is to be noted that toner dropped from the first trapping filter **621** due to vibrations would be deposited on the magnetic brush formed on the outer circumferential surface of the

developing roller **44**.
(80) Alternatively, the vibration generating part **14** may be provided in the developing device **40**. In that case, preferably, size and output of the vibrating motor are adjusted, as the case may be, in accordance with a placement site of the vibration generating part **14**.
(81) Now, evaluation of toner scattering in the image forming apparatus **1** will be described below. In this evaluation, printing with images at a print coverage rate of 5% per sheet S was carried out in a total of 5,000 sheets on a basis of 1,000 sheets for each of one-sheet intermittence (one-second halt after every one-sheet printing), five-sheet intermittence (one-second halt after every five-sheet printing; ditto for the followings), ten-sheet intermittence, twenty-sheet intermittence, and continuous printing. Each time 5,000 sheets were printed, the toner collection mode was executed. Then, printing in the same manner was carried out up to 100,000 sheets, and toner scattering onto the top surface of the development container **50** was checked.
(82) In execution of intermittent printing, there is a need for driving the developing device **40** during a certain stabilization period (e.g., 5 sec) before and after printing in order to obtain system stabilization. In cases of smaller-number-of-sheets intermittent printing such as one-sheet intermittence and five-sheet intermittence, when an identical number of sheets (e.g., 1,000 sheets) are printed, the number of times of the stabilization period increases (1,000 times for one-sheet intermittence, 200 times for five-sheet intermittence), hence a fear for generation of larger quantities of toner scattering. For this reason, system speed is set lower than normal in smaller-number-of-sheets intermittent printing. Since toner scattering decreases more and more with lowering system speed, generation of toner scattering can be suppressed in smaller-number-of-sheets intermittent printing such as one-sheet intermittence and five-sheet intermittence. In particular, in the case of one-sheet intermittence, further lowering the system speed than in the case of five-sheet intermittence allows the generation of toner scattering to be suppressed to more extent.
(83) The image forming apparatus **1** used in this evaluation is that of Modification Example 2 described above with reference to FIGS. **8** and **9**. That is, the development container **50** of the image forming apparatus **1** in this Example includes the air vent **56** and the discharge part **58**. In contrast to this, the image forming apparatus of Comparative Example has no air vent provided in the development container.
(84) Then, as to the evaluation of toner scattering, first evaluation and second evaluation differing in rotational conditions for the developing roller from each other were carried out.
(85) In connection with the first evaluation, evaluation conditions and evaluation results of Example 1 are shown in Table 1, while evaluation conditions and evaluation results of Comparative Example 1 are shown in Table 2.
(86) TABLE-US-00001 TABLE 1 UNDER TONER EXAMPLE 1 UNDER VIBRATIONS COLLECTION VIBRATION ON OFF GENERATING PART DEVELOPING HALT REVERSE ROLLER ROTATION FILTER TONER TONER FAN RECOVERABILITY SCATTERING SCATTERING SUCTION SLIGHTLY FAULTY NO SLIGHT HALT SUCCESSFUL SLIGHT PARTIAL DISCHARGE VERY SUCCESSFUL PARTIAL PARTIAL
(87) TABLE-US-00002 TABLE 2 COMPARATIVE UNDER TONER EXAMPLE 1 UNDER VIBRATIONS COLLECTION VIBRATION ON OFF GENERATING PART DEVELOPING HALT REVERSE ROLLER ROTATION FILTER TONER TONER FAN RECOVERABILITY SCATTERING SCATTERING SUCTION SLIGHTLY FAULTY SLIGHT LARGE- QUANTITY HALT SUCCESSFUL PARTIAL VERY LARGE- QUANTITY DISCHARGE VERY SUCCESSFUL PARTIAL VERY LARGE- QUANTITY
(88) The developing roller **44** was set to a halt of rotation under a state of ‘vibrations’ with the vibration generating part **14** held drive-on, and the developing roller **44** was set to reverse rotation (clockwise in FIG. **4**) under a state of ‘toner collection’ with the vibration generating part **14** held drive-off. Toner trapped by the trapping filter **62** is dropped onto the magnetic brush formed on the

outer circumferential surface of the developing roller **44** under a state of 'vibrations'. Further, under a state of 'toner collection', the developing roller **44**, to which the developing voltage is applied, is set to reverse rotation, the dropped toner is collected via the photosensitive drum **21** in the drum cleaning part **23**.
(89) Then, in each case of 'under vibrations' and 'under toner collection', the fan **63** of the toner trapping mechanism **60** was driven under conditions of 'suction', 'halt' and 'discharge'. Under 'suction', the fan **63** is rotated forward so that air within the second conveyance chamber **53** is sucked into the duct **61**. For this reason, the recoverability of the trapping filter **62** is slightly faulty. Under 'halt', rotation of the fan **63** is halted. For this reason, the recoverability of the trapping filter **62** is successful. Under 'discharge', making the fan **63** rotated reverse allows toner trapped by the trapping filter **62** to be discharged from within the duct **61** to the second conveyance chamber **53** side. For this reason, the recoverability of the trapping filter **62** is very successful.
(90) According to Table 2, with the image forming apparatus of Comparative Example 1, under 'vibrations', suction of the fan, when executed, led to slight quantity of toner scattering onto the top surface of the development container. On the other hand, in the cases with the fan under halt and under discharge, toner scattering occurred on part of the top surface of the development container. Furthermore, under 'toner collection', suction of the fan, when executed, led to large-quantity of toner scattering occurrence, and halt and discharge of the fan, when executed, led to very large-quantity of toner scattering occurrence.
(91) In contrast to this, according to Table 1, with the image forming apparatus **1** of Example 1 of the present disclosure, when the fan **63** was set to 'discharge' under 'vibrations', toner scattering occurred in part of the top surface of the development container **50**, yet toner scattering became slight with the fan **63** set to halt, and no toner scattering occurred with the fan **63** set to suction. Further, under toner collection, with the fan **63** set to halt and discharge, toner scattering occurred on part of the top surface of the development container **50**, while with the fan **63** set to suction, slight quantity of toner scattering occurred.
(92) In connection with the second evaluation, evaluation conditions and evaluation results of Example 2 are shown in Table 3, and evaluation conditions and evaluation results of Comparative Example 2 are shown in Table 4.
(93) TABLE-US-00003 TABLE 3 UNDER TONER EXAMPLE 2 UNDER VIBRATIONS COLLECTION VIBRATION ON OFF GENERATING PART DEVELOPING FORWARD ROTATION ROLLER FILTER TONER TONER FAN RECOVERABILITY SCATTERING SCATTERING SUCTION SLIGHTLY FAULTY SLIGHT NO HALT SUCCESSFUL PARTIAL SLIGHT DISCHARGE VERY SUCCESSFUL PARTIAL PARTIAL
(94) TABLE-US-00004 TABLE 4 COMPARATIVE UNDER TONER EXAMPLE 2 UNDER VIBRATIONS COLLECTION VIBRATION ON OFF GENERATING PART DEVELOPING FORWARD ROTATION ROLLER FILTER TONER TONER FAN RECOVERABILITY SCATTERING SCATTERING SUCTION SLIGHTLY FAULTY LARGE- PARTIAL QUANTITY HALT SUCCESSFUL LARGE- LARGE- QUANTITY QUANTITY DISCHARGE VERY SUCCESSFUL VERY LARGE- LARGE- QUANTITY QUANTITY
(95) The developing roller **44** was set to forward rotation (counterclockwise in FIG. **4**) under a state of 'vibrations' with the vibration generating part **14** held drive-on, and the developing roller **44** was set to continue forward rotation as it was under a state of 'toner collection' with the vibration generating part **14** held drive-off. Toner trapped by the trapping filter **62** is dropped onto the magnetic brush formed on the outer circumferential surface of the developing roller **44** under a state of 'vibrations'. Further, under a state of 'toner collection', the developing roller **44**, to which the developing voltage is applied, is set to forward rotation, and the dropped toner is collected to lower portion within the second conveyance chamber **53** and reused.
(96) Drive conditions for the fan **63** of the toner trapping mechanism **60** are the same as in the foregoing first evaluation.

(97) According to Table 4, with the image forming apparatus of Comparative Example 2, under 'vibrations', suction and halt of the fan, when executed, led to occurrence of large-quantity toner scattering onto the top surface of the development container, and discharge of the fan, when executed, led to occurrence of very large-quantity toner scattering. Further, under 'toner collection', suction of the fan, when executed, led to occurrence of toner scattering on part of the top surface of the development container, and halt and discharge of the fan, when executed, led to occurrence of large-quantity toner scattering.
(98) In contrast to this, according to Table 3, with the image forming apparatus **1** of Example 2 of the present disclosure, under 'vibrations', when the fan **63** was set to 'halt' and 'discharge', toner scattering occurred in part of the top surface of the development container **50**, but toner scattering became slight with the fan **63** set to suction. Further, under 'toner collection', toner scattering occurred in part of the top surface of the development container **50** with the fan **63** set to discharge, but toner scattering became slight with the fan **63** set to halt, and no toner scattering occurred with the fan **63** set to suction.
(99) As described above, according to the configuration of this embodiment, it can be considered that suppression of pressure increases within the first conveyance chamber **52** has been realized effectively, trapping of more than necessary quantity of toner by the trapping filter **62** has been suppressed, and thus trapping-performance stability of the toner trapping mechanism **60** has been improved. As a result of this, it can be concluded that toner scattering onto the top surface of the development container **50** has been suppressed.
(100) FIG. **10** is a vertical cross-sectional front view of the developing device **40** according to Modification Example 3. The developing device **40** of Modification Example 3 includes a film member **46**.
(101) The film member **46** is set onto an inner wall of the development container **50** in proximity to the air inlet port **611** of the toner trapping mechanism **60**. In more detail, the film member **46** is mounted on an inner wall of the development container **50** downstream of the air inlet port **611** as viewed in a rotational direction (forward rotation, counterclockwise in FIG. **10**) of the developing roller **44** during image formation. The film member **46** extends over an entire length of the developing roller **44** in its axial direction. The film member **46** extends toward the developing roller **44**.
(102) Then, the film member **46** is placed in contacting opposition to the outer circumferential surface of the developing roller **44**. In more detail, out of an elongation-forward end portion of the film member **46** as elongated from its mounting site, a planar portion of the film member **46** is put into contact with the outer circumferential surface of the developing roller **44**.
(103) Next, in terms of presence or absence of the film member **46**, evaluation of toner scattering inside the image forming apparatus **1** is described below. It is to be noted that various conditions in this evaluation are the same as those described with reference to Tables 1, 2, 3 and 4 and therefore description thereof is omitted here. In addition, drive conditions for the fan **63** of the toner trapping mechanism **60** were assumed as the only case of 'suction'.
(104) The image forming apparatus **1** of Example 1 used in this evaluation is that described above with reference to Table 1. The developing device **40** of Example 1 includes no film member **46**. By contrast, the image forming apparatus **1** of Example 3 includes the film member **46** shown in FIG. **10**. Evaluation conditions and evaluation results are shown in Table 5.
(105) TABLE-US-00005 TABLE 5 UNDER TONER UNDER VIBRATIONS COLLECTION VIBRATION ON OFF GENERATING PART DEVELOPING HALT REVERSE ROLLER ROTATION FILTER TONER TONER RECOVERABILITY SCATTERING SCATTERING EXAMPLE 1 SLIGHTLY FAULTY NO SLIGHT EXAMPLE 3 SLIGHTLY FAULTY NO NO
(106) According to Table 5, with the image forming apparatus **1** of Example 1 of the present disclosure, there occurred no toner scattering onto the top surface of the development container **50** 'under vibrations', and slight toner scattering occurred 'under toner collection'. By contrast, with

the image forming apparatus **1** of Example 3 of the present disclosure, there occurred no toner scattering onto the top surface of the development container ‘under vibrations’, neither toner scattering occurred also ‘under toner collection’.

(107) As described above, according to the configuration of Example 3, by virtue of the film member **46** being included therein, toner present in developer-stirring regions in which the first conveyance member **42** and the second conveyance member **43** are placed can be made less likely to reach the toner trapping mechanism **60** with an enhanced effect. That is, the suppressing effect for the occurrence that more than necessary quantity of toner is trapped by the trapping filter **62** can be enhanced, so that the stability in trapping performance of the toner trapping mechanism **60** can be enhanced to even more extent. Thus, as ascertained by the above results, it becomes implementable to suppress toner scattering onto the top surface of the development container **50**.

(108) Although an embodiment of the present disclosure has been described hereinabove, yet the scope of the disclosure is not limited to this, and the disclosure may be changed and modified in various ways unless those changes and modifications depart from the gist of the disclosure.

(109) For example, in the above-described embodiment, the image forming apparatus **1** is assumed as being a tandem-type image forming apparatus for color printing in which images of plural colors are formed in successive superimposition on one another. However, the image forming apparatus is not limited to such models. The image forming apparatus may be a non-tandem-type image forming apparatus for color printing or an image forming apparatus for monochrome printing.

Claims

1. A developing device comprising: a development container including a first conveyance chamber and a second conveyance chamber placed in parallel juxtaposition and a first communicating portion and a second communicating portion through which the first and second conveyance chambers communicate with each other at their longitudinal both end portions, the development container containing a toner-containing developer to be fed to an image carrier; a first conveyance member and a second conveyance member which are rotatably supported by the first conveyance chamber and the second conveyance chamber, respectively, and which convey and circulate the developer, while stirring the developer, in mutually opposite directions of the longitudinal direction; a developer carrier which is rotatably supported by the development container in opposition to the image carrier, and which feeds the toner contained in the second conveyance chamber to the image carrier; and a toner trapping mechanism for trapping the toner contained in the second conveyance chamber, wherein the toner trapping mechanism includes: a duct which is connected to the second conveyance chamber and which allows air present in the second conveyance chamber to flow therethrough; an air inlet port which is opened at a connecting portion between the duct and the second conveyance chamber and which allows air present in the second conveyance chamber to flow into the duct; a trapping filter which traps the toner contained in air flowing through the duct; and a fan which allows air present in the second conveyance chamber to be sucked into the duct and let to flow outside, the developer is a two-component developer containing magnetic carrier and the toner, the development container has: an air vent disposed at a location different from the first and second communicating portions, the air vent being opened in a circumferential wall part of the first conveyance chamber from inside to outside the air vent being communicated with outside of the development container; and a developer discharge port which is formed at a downstream-side end portion of the second conveyance chamber as viewed in a developer conveyance direction of the second conveyance chamber and which allows excess portion of the developer present within the development container to be discharged outside, and the air vent is placed on a downstream-side end portion of the first conveyance chamber in its developer conveyance direction, the downstream-side end portion being opposite to a side on which the developer discharge port is placed.

2. The developing device according to claim 1, wherein the air vent is formed at an upper portion of the first conveyance chamber.
3. The developing device according to claim 1, wherein the development container includes a ventilation filter which covers the air vent.
4. The developing device according to claim 1, wherein the air vent is placed at at least a downstream-side end portion out of both upstream- and downstream-side end portions of the first conveyance chamber as viewed in a developer conveyance direction of the first conveyance chamber.
5. The developing device according to claim 1, wherein the trapping filter includes: a first trapping filter which covers the air inlet port; and a second trapping filter which is placed downstream of the first trapping filter in the duct, as viewed in an airflow direction of the duct, to cover an airflow cross section and which is higher in toner-trapping efficiency than the first trapping filter.
6. The developing device according to claim 1, further comprising a film member which is attached to an inner-wall portion of the development container downstream of the air inlet port as viewed in a rotational direction of the developer carrier during image formation, the film member extending toward the developer carrier so as to be placed in contacting opposition to an outer circumferential surface of the developer carrier, and in an elongation-forward end portion of the film member as elongated from its mounting site, a planar portion of the film member makes contact with the outer circumferential surface of the developer carrier.
7. An image forming apparatus including the developing device according to claim 1.
8. The image forming apparatus according to claim 7, further comprising a vibration generating part including a vibrating motor and an exciting weight attached on an output shaft of the vibrating motor, the vibration generating part vibrating the trapping filter.
